EARTHPRESS · CONCEPT OPTIONS

EarthPress concept options

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

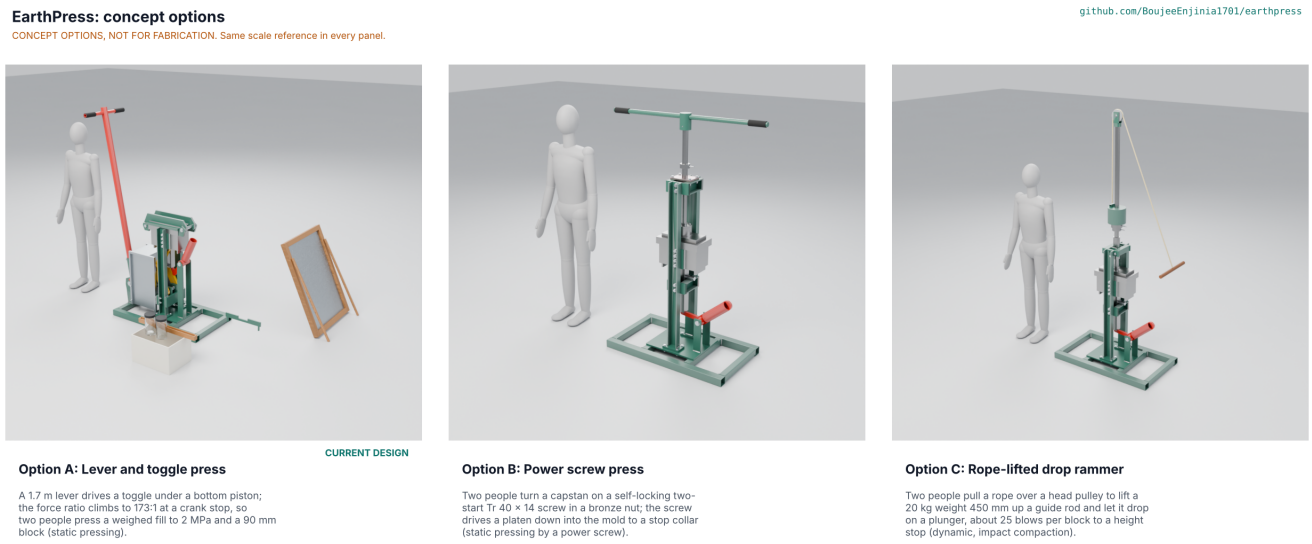
Document ID	EPR-OPT-001	Version	0.1
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/earthpress		
Source	docs/02a-concept-options.md @ 212cbb1		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First concept options (kit 1.8 pipeline, kit 1.9.1 detailed models); the current design is option A; scored against EPR-URS-001; all three within 10 %, sent to Amish without a pick

EarthPress concept options

No recommendation: the current lever and toggle press (option A) scores 186 of 245, the power screw press (option B) 181 and the rope-lifted drop rammer (option C) 168. B is 2.7 % and C 9.7 % below A, so under STANDARDS section 21 all three go to Amish without a pick, recorded as "Proposed, awaiting Amish" (EPR-DEC-001 D1, EPR-DDR-004). The repo stays on option A, the design it was built around, unless Amish decides otherwise. Every option is modelled at detail level D3 in `cad/src/model.py` and `cad/src/epr_options.py` (STANDARDS section 24).



The options picture is rendered on Amish's Mac with `python .kit/options.py` (Blender Cycles); until then this link has no image.

The options

Option A: lever and toggle press (the current design)

A 1.7 m lever drives a toggle under a bottom piston; the force ratio climbs to 173:1 at a crank stop, so two people press a weighed fill to 2 MPa and a 90 mm block. This is the constructable design of EPR-DDR-003 with fix F1 (40 mm 42CrMo4 +QT toggle pins): a welded base on skids, a bolted press core of UPN 80 columns, a 12 mm plate mold with a ribbed, pinned lid, a slotted push rod, four toggle links in bushes, a lever hub with crank, catch and pawl, an eject seesaw and a linkage guard. It is the CINVA-Ram pattern with every member sized (EPR-CAL-001).

- First-order numbers: 81.2 kN at 2 MPa; peak pull 405 N per operator for the softer design soil; about 296 blocks a day with a crew of four; 203.8 kg in 11 pieces of 44.5 kg or less; USD 542 in parts.
- Key risks: the narrow fill window (0 to +2.7 %, weighed fills), 47 J of kickback held by the end pawl, the soil compaction law not yet measured (EPR-STM-001 S6).
- Needs from a co-design partner: soils to measure the compaction curve, acceptance of a two-person lever, timed cycles.

Option B: power screw press

Two people turn a capstan on a self-locking two-start Tr 40 x 14 screw in a bronze nut; the screw drives a platen down into the mold to a stop collar. An H frame of UPN 80 columns, a bottom beam and a crosshead carry the nut; a removable stop pin holds a floor piston during pressing, and an eject seesaw lifts it to push the block out.

- First-order numbers: 406 N per operator on 0.53 m capstan grips at 81.2 kN (screw efficiency about 43 %); about 225 blocks a day (111 s cycle, 14 unloaded turns each way); 165.5 kg in the model; about USD 460 plus the screw, bronze nut and thrust bearing (not priced, to assess).
- Key risks: a greased two-start screw may not self-lock (EPR-STM-001 B2), so the platen could run back; thread wear in gritty soil; the bought screw, nut and bearing are specialist parts.
- Needs from a co-design partner: a source for trapezoidal screws and bronze nuts; acceptance of a slower cycle.

Option C: rope-lifted drop rammer

Two people pull a rope over a head pulley to lift a 20 kg sleeve weight 450 mm up a guide rod and let it fall on an anvil collar; the plunger on the rod compacts the soil with about 25 blows to a height stop. The same base, columns, mold, floor piston and eject seesaw as B, with a light crosshead and guide sleeve instead of the screw.

- First-order numbers: 25 blows of 88 J give standard Proctor energy for the block (2.2 kJ); 98 N per person on the rope (196 N when lifting the plunger for filling); about 160 blocks a day (160 s cycle); 170.5 kg in the model; about USD 470 plus the pulley and rope (not priced, to assess).
- Key risks: impact compaction densifies the top of the block most and its equivalence with 2 MPa static pressing is unproven (C2); a falling weight about 4,000 times a day; the impact force on the mold and floor stop is unknown (C6).
- Needs from a co-design partner: test blocks to compare density and strength with pressed blocks.

First-order numbers

Table 1. Options compared (mass and parts count from `cad/fidelity.json`; A's mass is the press without the soil kit, from EPR-CAL-001 section 8).

Quantity	Option A	Option B	Option C
Estimated parts cost (USD)	542 (priced BOM)	About 460 plus screw, nut and bearing (to assess)	About 470 plus pulley and rope (to assess)
Mass (kg)	203.8 (press)	165.5	170.5
Output or throughput	About 296 blocks a day	About 225 blocks a day	About 160 blocks a day
Energy or effort	405 N per operator at the peak, for a few seconds	406 N per operator over 4.3 loaded turns, plus 28 light turns	98 N per person, 25 pulls per block
Parts count (D3 model)	251 (with the soil kit)	185	165
STEM check	Within, after fix F1; open on S2, S6, S18	Plausible on self-locking (B2) and nut bearing (B3)	Plausible on density (C2), output (C5) and impact force (C6)

Costs for B and C are first-order: the soil kit, consumables and fasteners as A (USD 113), plus the rest at A's cost per kilogram of press (USD 2.10/kg, steel, cutting and the bought pins and bushes). They are not priced from quotes.

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2, Low 1; scores 1 to 5; URs from EPR-URS-001).

Criterion	Source	Weight	A	B	C	Reason
Standard block size and height	UR-001	3	5	4	3	A's crank stop fixes 90 mm; B's stop collar fixes it but the frame stretches under load; C's stop sets height but density varies top to bottom
Dense, strong blocks (2 MPa)	UR-002	3	4	5	2	A reaches 2 MPa within a weighed fill window; B has pressure in hand at any fill; C's equivalence with pressing is unproven (C2)
Output for a house	UR-003	3	4	3	2	296, about 225 and about 160 blocks a day against 300
Ordinary adults can operate it	UR-004	3	4	4	3	A and B need about 405 N each; C's pulls are light but about 4,000 a day per pair
Built and repaired by a local welder	UR-005	3	4	3	5	A needs bought turned pins and bushes; B a trapezoidal screw, bronze nut and thrust bearing; C only plate, tube, bar, a pulley and rope
Movable between sites	UR-006	2	3	4	4	203.8, 165.5 and 170.5 kg
Long life, replaceable wear parts	UR-007	2	3	2	4	A has ten bushes and grease points; B's screw and nut wear in grit; C has few moving parts and a consumable rope
Mix chosen with field tests	UR-008	3	4	4	4	Same soil kit for all
Affordable press and kit	UR-009	3	3	3	4	A USD 542; B and C below A before their bought parts are priced
Low stabilizer use	UR-010	2	4	5	3	B's pressure margin could let cement fall; C's uncertain density may need more
Lower carbon than fired brick	UR-011	2	4	4	3	Same blocks for A and B; C may need more cement
Safe for operators and bystanders	UR-012	3	3	4	2	A stores 47 J at the stop (pawl, guard); B stores little if it self-locks; C drops 20 kg about 4,000 times a day
Open documentation	UR-013	2	4	4	4	Same documentation route
Human powered	UR-014	3	5	5	5	None uses fuel, electricity or hydraulics
Works outdoors in dust and rain	UR-015	2	4	2	4	A's bushes have dust seals; B's threads are exposed; C's rod and collars tolerate grit
Blocks testable against a strength reference	UR-016	2	4	4	3	A and B make uniform pressed blocks; C's density gradient widens test scatter
Cost against the value-engineering target	STANDARDS s21	2	3	3	4	A is USD 42 over; B and C may come in under
Buildable with local skills and tools	STANDARDS s21	2	4	3	5	As UR-005
Safety		3	3	4	2	As UR-012

Criterion	Source	Weight	A	B	C	Reason
	STANDARDS s21					
Maintenance	STANDARDS s21	1	3	2	4	As UR-007
Weighted total (of 245)			**186**	**181**	**168**	A first; B 2.7 % and C 9.7 % below A

Recommendation

None. The three options are within 10 % of each other, so they go to Amish without a pick (STANDARDS section 21), as decision D1 in EPR-DEC-001, argued in EPR-DDR-004. The repo stays on option A, the existing design, unless he chooses otherwise; changing to B or C would change the pitch ("a lever and toggle linkage") and is his decision.

Ideas from the runners-up worth grafting onto A whichever is chosen: from B, a positive stop that does not depend on the operators (A already has one in its crank stop) and a capstan-like long arc for a single operator, which would need a lower force ratio; from C, the simplicity of its parts list, which argues for keeping A's bought parts to the pins, bushes and springs.